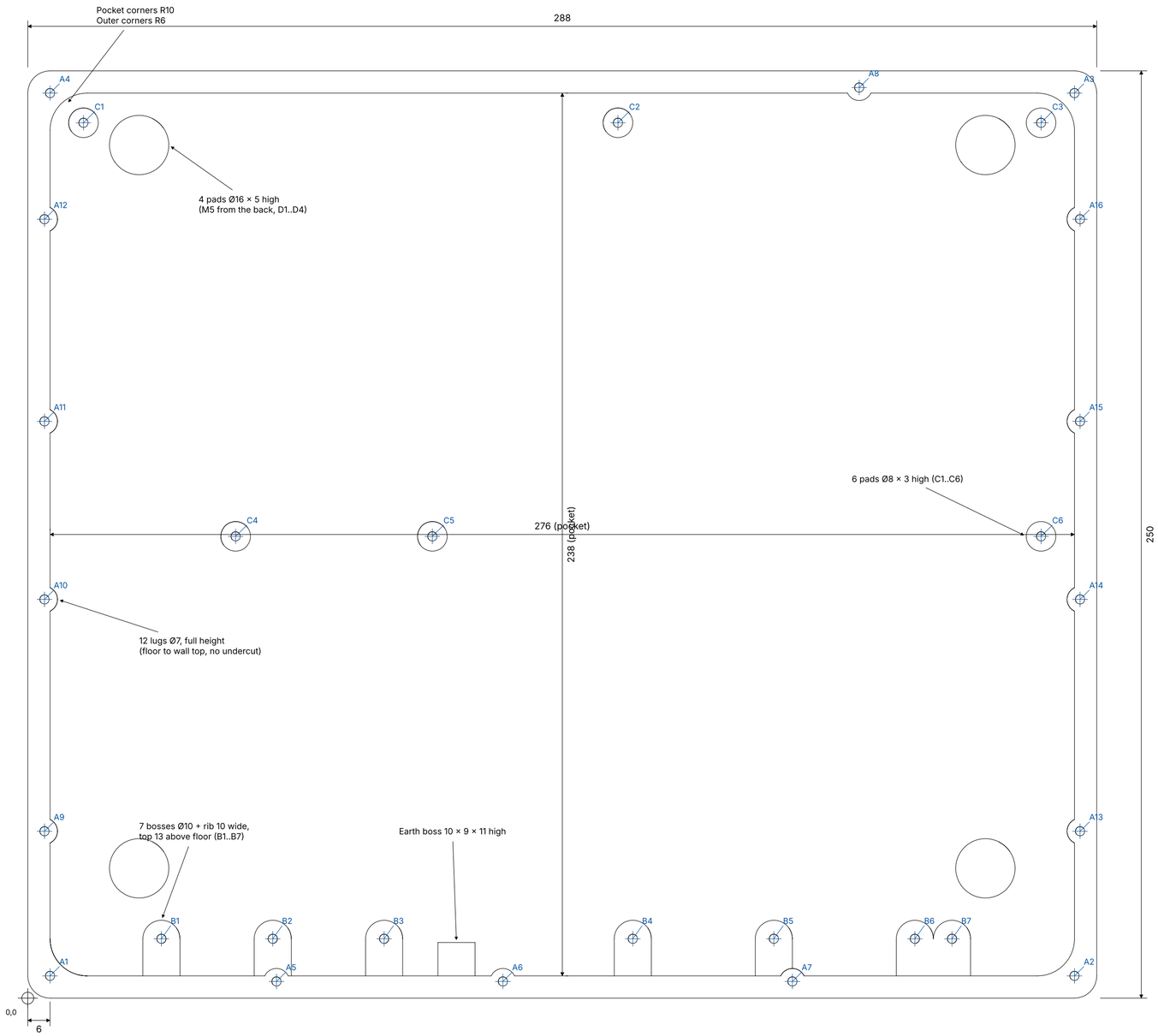
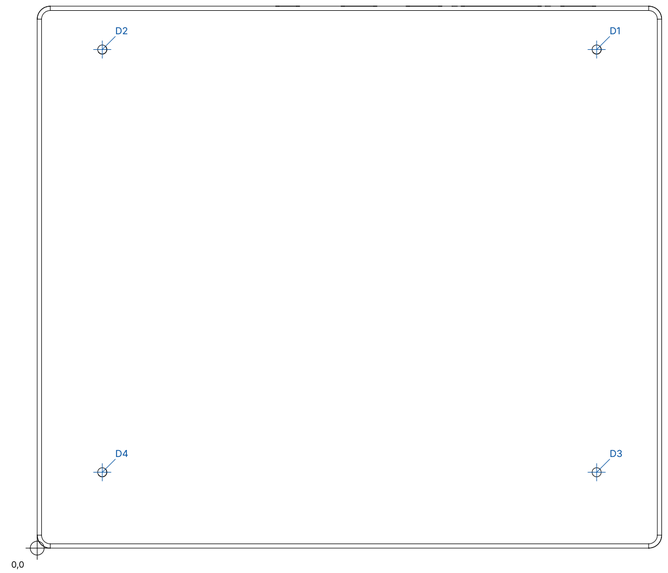


TRAY - PLAN (inside, seen from the lid side)

Floor 5 thick, walls 6, tray height 63. Tapped holes A, B, C: positions in the hole table (sheet 6), datum 0,0 = outer corner shown.

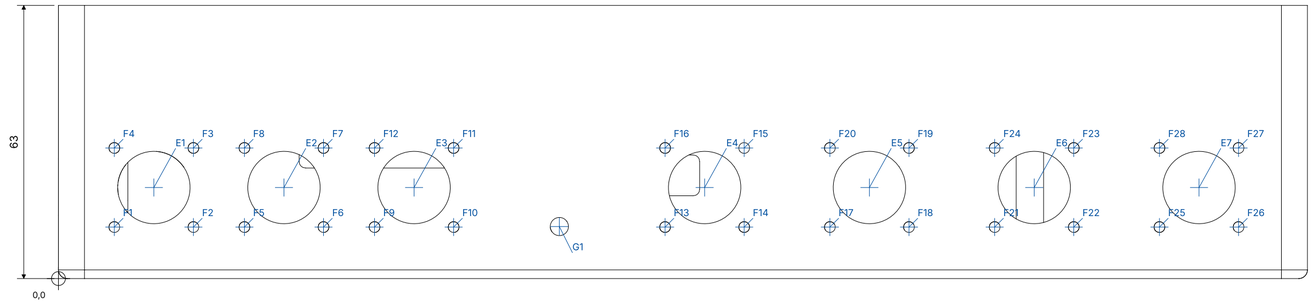


TRAY - BACK FACE (seen from the back plate)

Seen from the back: case x runs right to left. 4 × M5 (D1..D4) into the inside pads.

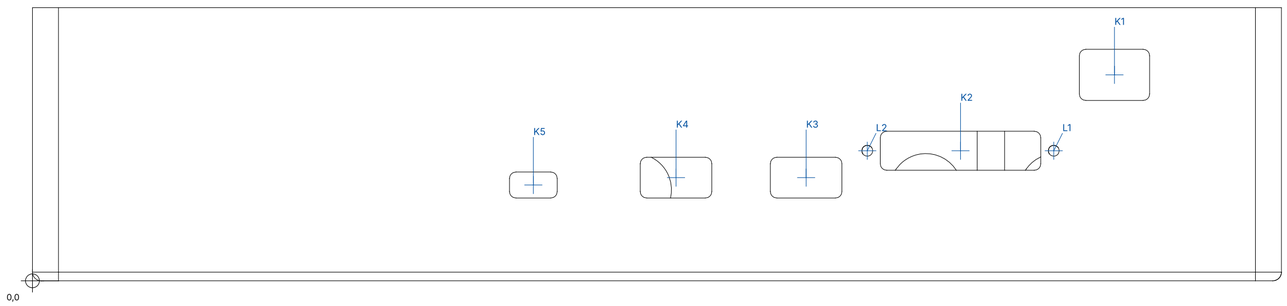
Back edge R2 all round. Flat face: mounts on the back plate via printed spacers.

Shack Relay Matrix rev C - RF unit case	
Tray: plan (inside) and back face	Dwg RF-CASE-01 rev A
Material EN AW-6061-T6 (6082-T6 accepted). Finish: conductive conversion coating (JLC 'Conductive anodizing'), MIL-DTL-5541 Type III/Cl. 3	Sheet 1/7 plan 1:1, back 1:2
Tolerances ISO 2768-mK. Dimensions in mm. Third angle projection. Master geometry: rf-case.step	2026-10-24 PE1RGU (P. Doornekamp)



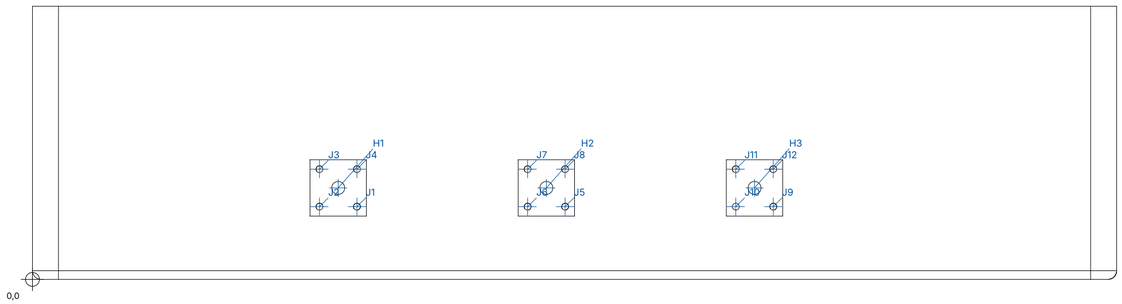
WALL 1 - SO-239 WALL (outside)

E: Ø16.7 THRU, flange outside, 4 × M3 on 18.24 square. G: earth strap M5 (tinned braid to the RVS earth bar).



WALL 2 - POWER / RJ45 WALL (outside)

K: cutouts THRU, corner R1.5, sizes in the hole table. EXP plate (sheet 5) covers K2 with 2 × M3 (L).



WALL 3 - SMA WALL (outside)

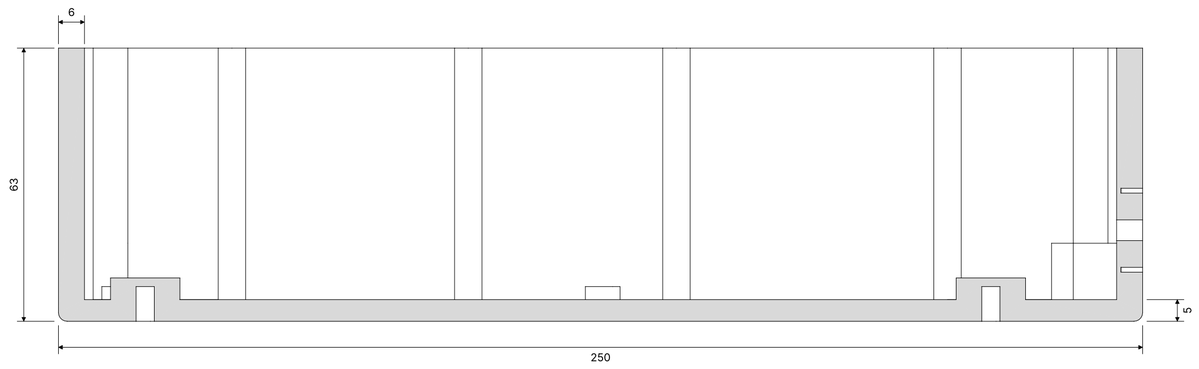
H: pocket 13 × 13 × 2 deep from outside (flange seat 12.7 square, corner R ≤ 1 allowed), Ø3.0 THRU at centre; J: 4 × M2 in the pocket floor.



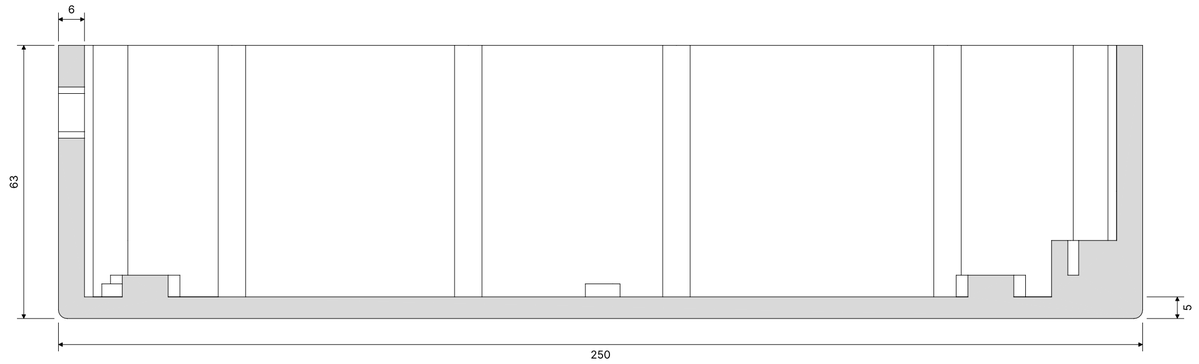
WALL 4 - PLAIN WALL (outside)

Wall 4 has no openings; the 4 side lugs are inside only.

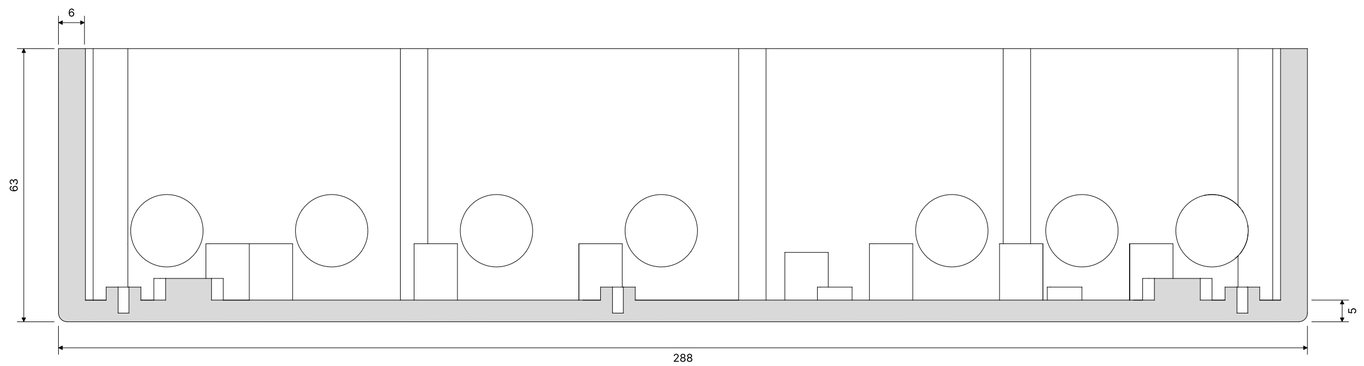
Shack Relay Matrix rev C - RF unit case	
Tray: the four walls, outside views	Dwg RF-CASE-01 rev A
Material EN AW-6061-T6 (6082-T6 accepted). Finish: conductive conversion coating (JLC 'Conductive anodizing'), MIL-DTL-5541 Type 2, Cl 3.	Sheet 2 / 7 3.1:1
Tolerances ISO 2768-mK. Dimensions in mm. Third angle projection. Master geometry: rf-case.step	2026-10-24 PE1RGU (P. Doornekamp)



SECTION A-A (x = 30: mounting pads, floor, walls)



SECTION B-B (x = 36: GND boss B1 with rib, SO-239 wall)



SECTION C-C (y = 236: standoff pads S1..S3, lugs)

Section faces shaded.

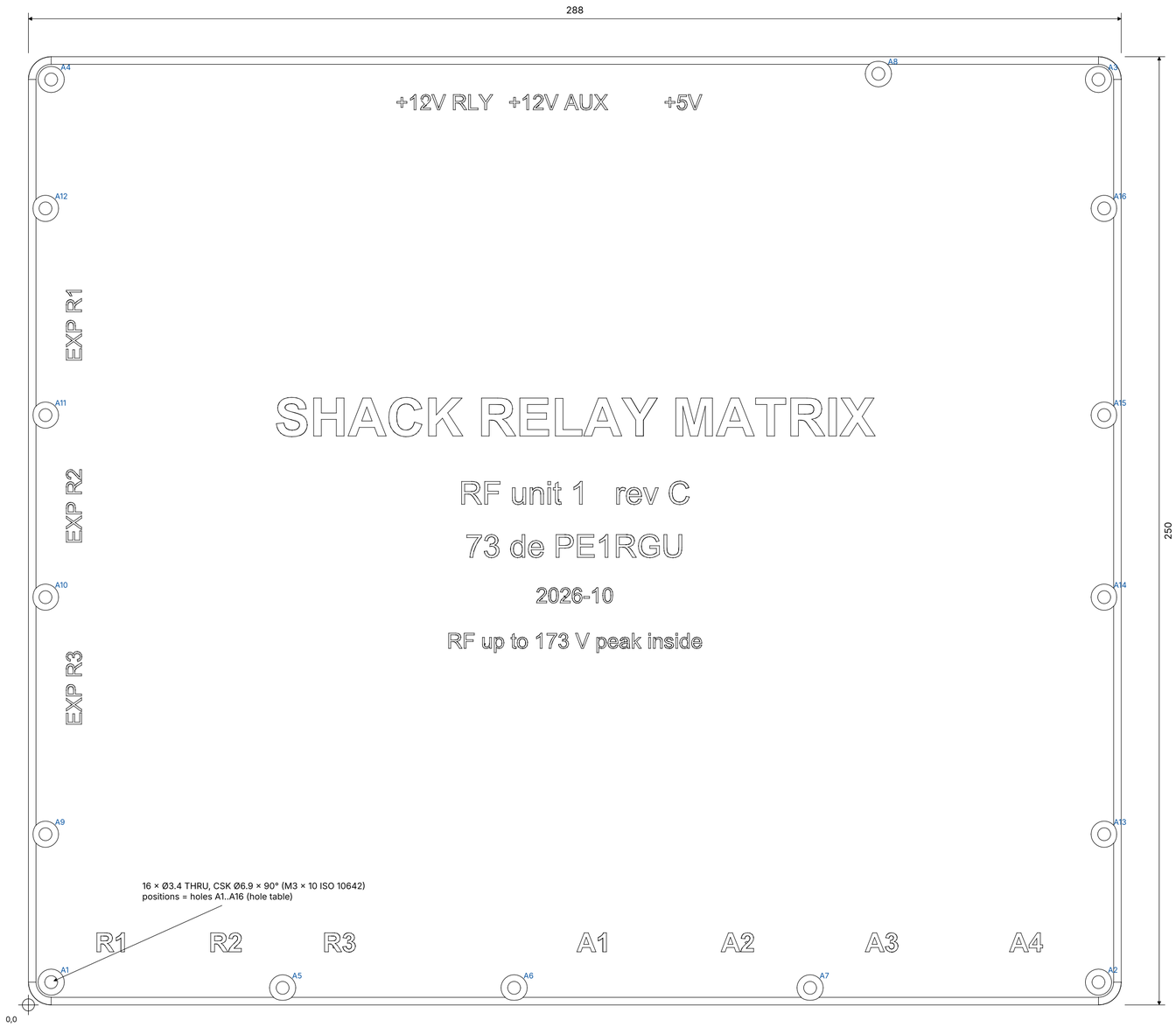
Floor 5, walls 6, lugs Ø7 full height to the wall top.

Mount pads 5 high (floor + pad = 10 for M5 × 8).

Standoff pads 3 high; GND bosses 13 high (board seat).

Concave vertical edges lug/boss to wall: R ≤ 2 allowed.

Shack Relay Matrix rev C - RF unit case	
Tray: sections	Dwg RF-CASE-01 rev A
Material EN AW-6061-T6 (6082-T6 accepted). Finish: conductive conversion coating (JLC 'Conductive anodizing'), MIL-DTL-5541 type 3, cl 3.1.1	Sheet 3 / 7
Tolerances ISO 2768-mK. Dimensions in mm. Third angle projection. Master geometry: rf-case.step	2026-10-24 PE1RGU (P. Doornekamp)



LID - TOP (outside)

Lid 288 × 250 × 3, EN AW-6061-T6.

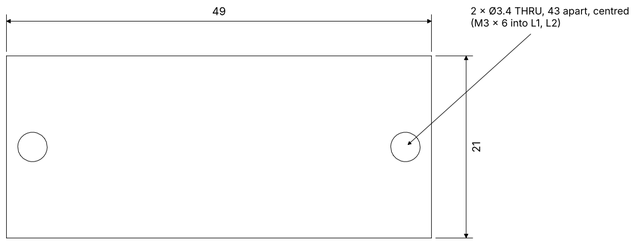
Outer corners R6, top edge R2, bottom edge sharp (seats on the tray).

Engraving 0.5 deep, Arial bold, as in rf-case.step (body Lid).

Optional: fill the engraving with white paint after coating.

Lid inside face and tray rim: no paint, conductive (RF bond).

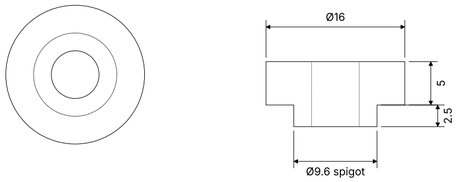
Shack Relay Matrix rev C - RF unit case	
Lid	Dwg RF-CASE-01 rev A
Material EN AW-6061-T6 (6082-T6 accepted). Finish: conductive conversion coating (JLC 'Conductive anodizing'), MIL-DTL-5541 Type II, Cl 3.	Sheet 1 / 1
Tolerances ISO 2768-mK. Dimensions in mm. Third angle projection. Master geometry: rf-case.step	2026-10-24 PE1RGU (P. Doornekamp)



EXP PLATE (2 mm, EN AW-6061-T6)

Unit 1: blank. Unit 2: ribbon clamp and shield contact (same outline, slot to suit). Same finish as the tray.

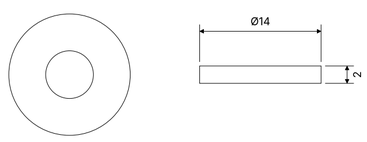
PRINTED SPACER (4×) 2:1



Bore Ø5.5 THRU. Spigot fits the Ø10 back-plate hole (plate 3 mm).

Flange side against the tray back, spigot into the plate.

PRINTED WASHER (4×) 2:1



Bore Ø5.5. Under the M5 × 16 head, behind the back plate.

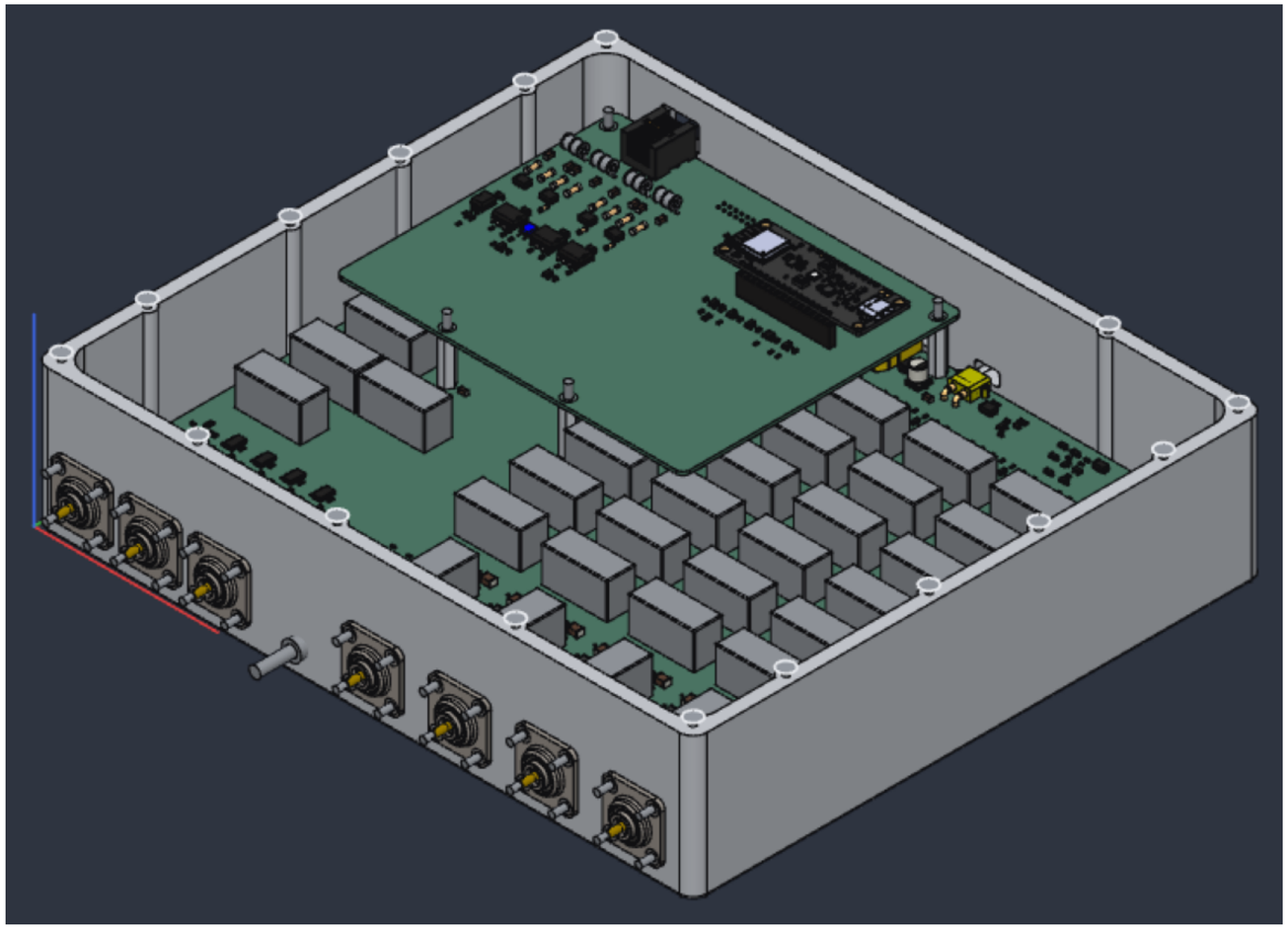
Print: PETG or ASA (better PA-CF), 100 % infill, layers parallel to the flange. Plastic creeps under bolt load:

low torque, spring washer under the head, check the four M5 after the first summer. Stack from the back:

M5 × 16 A4 + spring washer + printed washer | back plate | printed spacer | tray. The tray is isolated from the plate;

the only bond is the tinned braid on G1 (earth M5 in wall 1) to the RVS earth bar.

Shack Relay Matrix rev C - RF unit case	
EXP plate and printed mounting parts	Dwg RF-CASE-01 rev A
Material EN AW-6061-T6 (6082-T6 accepted). Finish: conductive conversion coating (JLC 'Conductive anodizing'), MIL-DTL-5541 Type II / Class 3.	Sheet 2 of 2
Tolerances ISO 2768-mK. Dimensions in mm. Third angle projection. Master geometry: rf-case.step	2026-10-24 PE1RGU (P. Doornekamp)



Assembly view (lid off): RF board on 6 standoffs, controller board on 4, SO-239 and SMA fitted.

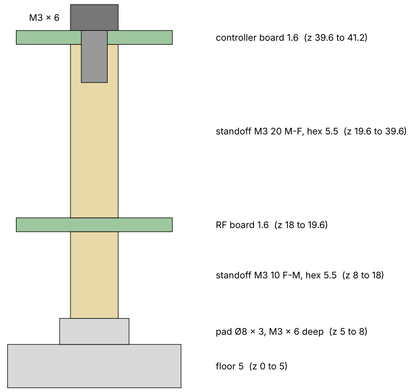
GND BOSS B1..B7: RF board seated on the boss top (z 18), M3 × 6 from the board top into M3 × 8.

LID: M3 × 10 countersunk ISO 10642 into A1..A16 (thread 8, lid 3, head flush with Ø6.9 countersink).

SO-239 83-798: flange outside wall 1, 4 x M3 × 6 into F (thread 5). SMA 901-9215: flange in pocket H, 4 x M2 × 4 into J.

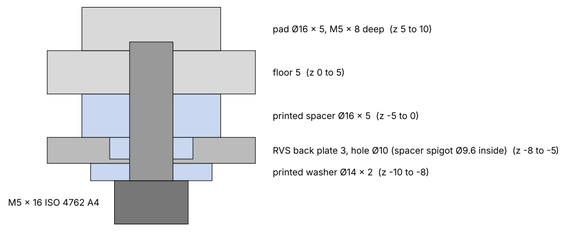
EXP plate: 2 x M3 × 6 into L. Earth: M5 × 16 with braid lug into G1 (thread 13).

Heights: RF board underside z 18, controller underside z 39.6, DevKit top about z 57.2, lid inside z 63.



STACK AT S1, S2, S4, S5 (2:1)

S3, S6: 10 mm standoff only, M3 × 6 on the RF board (z 19.6). Male studs: 6 mm into the thread below.



MOUNTING STACK D1..D4 (2:1)

Spring washer under the head. Tray isolated from the plate; bond only via G1.

Shack Relay Matrix rev C - RF unit case	
Assembly: stack-up, standoffs and fasteners	Dwg RF-CASE-01 rev A
Material EN AW-6061-T6 (6082-T6 accepted). Finish: conductive conversion coating (JLC 'Conductive anodizing'), MIL-DTL-5541 Type II, Class 3.	Sheet 2 of 3 details 2:1
Tolerances ISO 2768-mK. Dimensions in mm. Third angle projection. Master geometry: rf-case.step	2026-10-24 PE1RGU (P. Doornekamp)

ID	Feature	Size	Depth	View	X	Y	Note
A1	Lid screw	M3	8	1 plan	6.00	6.00	tap drill 2.5, max 10 deep
A2	Lid screw	M3	8	1 plan	282.00	6.00	tap drill 2.5, max 10 deep
A3	Lid screw	M3	8	1 plan	282.00	244.00	tap drill 2.5, max 10 deep
A4	Lid screw	M3	8	1 plan	6.00	244.00	tap drill 2.5, max 10 deep
A5	Lid screw	M3	8	1 plan	67.00	4.50	tap drill 2.5, max 10 deep
A6	Lid screw	M3	8	1 plan	128.00	4.50	tap drill 2.5, max 10 deep
A7	Lid screw	M3	8	1 plan	206.00	4.50	tap drill 2.5, max 10 deep
A8	Lid screw	M3	8	1 plan	224.00	245.50	tap drill 2.5, max 10 deep
A9	Lid screw	M3	8	1 plan	4.50	45.00	tap drill 2.5, max 10 deep
A10	Lid screw	M3	8	1 plan	4.50	107.50	tap drill 2.5, max 10 deep
A11	Lid screw	M3	8	1 plan	4.50	155.50	tap drill 2.5, max 10 deep
A12	Lid screw	M3	8	1 plan	4.50	210.00	tap drill 2.5, max 10 deep
A13	Lid screw	M3	8	1 plan	283.50	45.00	tap drill 2.5, max 10 deep
A14	Lid screw	M3	8	1 plan	283.50	107.50	tap drill 2.5, max 10 deep
A15	Lid screw	M3	8	1 plan	283.50	155.50	tap drill 2.5, max 10 deep
A16	Lid screw	M3	8	1 plan	283.50	210.00	tap drill 2.5, max 10 deep
B1	GND boss (board RF bond)	M3	8	1 plan	36.00	16.00	boss top = board seat
B2	GND boss (board RF bond)	M3	8	1 plan	66.00	16.00	boss top = board seat
B3	GND boss (board RF bond)	M3	8	1 plan	96.00	16.00	boss top = board seat
B4	GND boss (board RF bond)	M3	8	1 plan	163.00	16.00	boss top = board seat
B5	GND boss (board RF bond)	M3	8	1 plan	201.00	16.00	boss top = board seat
B6	GND boss (board RF bond)	M3	8	1 plan	239.00	16.00	boss top = board seat
B7	GND boss (board RF bond)	M3	8	1 plan	249.00	16.00	boss top = board seat
C1	Standoff pad S1..S6	M3	6	1 plan	15.00	236.00	tap drill 2.5, max 7 deep (floor stays 1 mm)
C2	Standoff pad S1..S6	M3	6	1 plan	159.00	236.00	tap drill 2.5, max 7 deep (floor stays 1 mm)
C3	Standoff pad S1..S6	M3	6	1 plan	273.00	236.00	tap drill 2.5, max 7 deep (floor stays 1 mm)
C4	Standoff pad S1..S6	M3	6	1 plan	56.00	124.50	tap drill 2.5, max 7 deep (floor stays 1 mm)
C5	Standoff pad S1..S6	M3	6	1 plan	109.00	124.50	tap drill 2.5, max 7 deep (floor stays 1 mm)
C6	Standoff pad S1..S6	M3	6	1 plan	273.00	124.50	tap drill 2.5, max 7 deep (floor stays 1 mm)
D1	Mounting, from the back	M5	8	1 back	258.00	230.00	tap drill 4.2, max 9 deep (0.5 mm must remain)
D2	Mounting, from the back	M5	8	1 back	30.00	230.00	tap drill 4.2, max 9 deep (0.5 mm must remain)
D3	Mounting, from the back	M5	8	1 back	258.00	35.00	tap drill 4.2, max 9 deep (0.5 mm must remain)
D4	Mounting, from the back	M5	8	1 back	30.00	35.00	tap drill 4.2, max 9 deep (0.5 mm must remain)
E1	SO-239 bore (Amphenol 83-798)	Ø16.7	THRU	2 wall 1	22.00	21.00	
E2	SO-239 bore (Amphenol 83-798)	Ø16.7	THRU	2 wall 1	52.00	21.00	
E3	SO-239 bore (Amphenol 83-798)	Ø16.7	THRU	2 wall 1	82.00	21.00	
E4	SO-239 bore (Amphenol 83-798)	Ø16.7	THRU	2 wall 1	149.00	21.00	
E5	SO-239 bore (Amphenol 83-798)	Ø16.7	THRU	2 wall 1	187.00	21.00	
E6	SO-239 bore (Amphenol 83-798)	Ø16.7	THRU	2 wall 1	225.00	21.00	
E7	SO-239 bore (Amphenol 83-798)	Ø16.7	THRU	2 wall 1	263.00	21.00	
F1	SO-239 flange screw	M3	5	2 wall 1	12.88	11.88	tap drill 2.5, max 5.5 deep (wall 6)
F2	SO-239 flange screw	M3	5	2 wall 1	31.12	11.88	tap drill 2.5, max 5.5 deep (wall 6)
F3	SO-239 flange screw	M3	5	2 wall 1	31.12	30.12	tap drill 2.5, max 5.5 deep (wall 6)
F4	SO-239 flange screw	M3	5	2 wall 1	12.88	30.12	tap drill 2.5, max 5.5 deep (wall 6)
F5	SO-239 flange screw	M3	5	2 wall 1	42.88	11.88	tap drill 2.5, max 5.5 deep (wall 6)
F6	SO-239 flange screw	M3	5	2 wall 1	61.12	11.88	tap drill 2.5, max 5.5 deep (wall 6)

ID	Feature	Size	Depth	View	X	Y	Note
F7	SO-239 flange screw	M3	5	2 wall 1	61.12	30.12	tap drill 2.5, max 5.5 deep (wall 6)
F8	SO-239 flange screw	M3	5	2 wall 1	42.88	30.12	tap drill 2.5, max 5.5 deep (wall 6)
F9	SO-239 flange screw	M3	5	2 wall 1	72.88	11.88	tap drill 2.5, max 5.5 deep (wall 6)
F10	SO-239 flange screw	M3	5	2 wall 1	91.12	11.88	tap drill 2.5, max 5.5 deep (wall 6)
F11	SO-239 flange screw	M3	5	2 wall 1	91.12	30.12	tap drill 2.5, max 5.5 deep (wall 6)
F12	SO-239 flange screw	M3	5	2 wall 1	72.88	30.12	tap drill 2.5, max 5.5 deep (wall 6)
F13	SO-239 flange screw	M3	5	2 wall 1	139.88	11.88	tap drill 2.5, max 5.5 deep (wall 6)
F14	SO-239 flange screw	M3	5	2 wall 1	158.12	11.88	tap drill 2.5, max 5.5 deep (wall 6)
F15	SO-239 flange screw	M3	5	2 wall 1	158.12	30.12	tap drill 2.5, max 5.5 deep (wall 6)
F16	SO-239 flange screw	M3	5	2 wall 1	139.88	30.12	tap drill 2.5, max 5.5 deep (wall 6)
F17	SO-239 flange screw	M3	5	2 wall 1	177.88	11.88	tap drill 2.5, max 5.5 deep (wall 6)
F18	SO-239 flange screw	M3	5	2 wall 1	196.12	11.88	tap drill 2.5, max 5.5 deep (wall 6)
F19	SO-239 flange screw	M3	5	2 wall 1	196.12	30.12	tap drill 2.5, max 5.5 deep (wall 6)
F20	SO-239 flange screw	M3	5	2 wall 1	177.88	30.12	tap drill 2.5, max 5.5 deep (wall 6)
F21	SO-239 flange screw	M3	5	2 wall 1	215.88	11.88	tap drill 2.5, max 5.5 deep (wall 6)
F22	SO-239 flange screw	M3	5	2 wall 1	234.12	11.88	tap drill 2.5, max 5.5 deep (wall 6)
F23	SO-239 flange screw	M3	5	2 wall 1	234.12	30.12	tap drill 2.5, max 5.5 deep (wall 6)
F24	SO-239 flange screw	M3	5	2 wall 1	215.88	30.12	tap drill 2.5, max 5.5 deep (wall 6)
F25	SO-239 flange screw	M3	5	2 wall 1	253.88	11.88	tap drill 2.5, max 5.5 deep (wall 6)
F26	SO-239 flange screw	M3	5	2 wall 1	272.12	11.88	tap drill 2.5, max 5.5 deep (wall 6)
F27	SO-239 flange screw	M3	5	2 wall 1	272.12	30.12	tap drill 2.5, max 5.5 deep (wall 6)
F28	SO-239 flange screw	M3	5	2 wall 1	253.88	30.12	tap drill 2.5, max 5.5 deep (wall 6)
G1	Earth strap	M5	13	2 wall 1	115.50	12.00	tap drill 4.2, max 14 deep (wall + boss 15)
H1	EXP SMA (901-9215): pocket 13 × 13 × 2 deep (corner R ≤ 1); centre bore	Ø3.0	THRU	2 wall 3	70.50	21.10	bore = 52 Ω air line around the Ø1.27 pin
H2	EXP SMA (901-9215): pocket 13 × 13 × 2 deep (corner R ≤ 1); centre bore	Ø3.0	THRU	2 wall 3	118.50	21.10	bore = 52 Ω air line around the Ø1.27 pin
H3	EXP SMA (901-9215): pocket 13 × 13 × 2 deep (corner R ≤ 1); centre bore	Ø3.0	THRU	2 wall 3	166.50	21.10	bore = 52 Ω air line around the Ø1.27 pin
J1	SMA flange screw (in pocket floor)	M2	3	2 wall 3	74.82	16.78	from pocket floor; tap drill 1.6, max 3.5 deep
J2	SMA flange screw (in pocket floor)	M2	3	2 wall 3	66.18	16.78	from pocket floor; tap drill 1.6, max 3.5 deep
J3	SMA flange screw (in pocket floor)	M2	3	2 wall 3	66.18	25.42	from pocket floor; tap drill 1.6, max 3.5 deep
J4	SMA flange screw (in pocket floor)	M2	3	2 wall 3	74.82	25.42	from pocket floor; tap drill 1.6, max 3.5 deep
J5	SMA flange screw (in pocket floor)	M2	3	2 wall 3	122.82	16.78	from pocket floor; tap drill 1.6, max 3.5 deep
J6	SMA flange screw (in pocket floor)	M2	3	2 wall 3	114.18	16.78	from pocket floor; tap drill 1.6, max 3.5 deep
J7	SMA flange screw (in pocket floor)	M2	3	2 wall 3	114.18	25.42	from pocket floor; tap drill 1.6, max 3.5 deep
J8	SMA flange screw (in pocket floor)	M2	3	2 wall 3	122.82	25.42	from pocket floor; tap drill 1.6, max 3.5 deep
J9	SMA flange screw (in pocket floor)	M2	3	2 wall 3	170.82	16.78	from pocket floor; tap drill 1.6, max 3.5 deep
J10	SMA flange screw (in pocket floor)	M2	3	2 wall 3	162.18	16.78	from pocket floor; tap drill 1.6, max 3.5 deep
J11	SMA flange screw (in pocket floor)	M2	3	2 wall 3	162.18	25.42	from pocket floor; tap drill 1.6, max 3.5 deep
J12	SMA flange screw (in pocket floor)	M2	3	2 wall 3	170.82	25.42	from pocket floor; tap drill 1.6, max 3.5 deep
K1	Cutout RJ45 J1, corners R1.5	16.2 × 11.8	THRU	2 wall 2	249.50	47.50	
K2	Cutout EXP ribbon slot, corners R1.5	37 × 9	THRU	2 wall 2	214.00	30.00	
K3	Cutout XT60 J11, corners R1.5	16.5 × 9.4	THRU	2 wall 2	178.40	23.80	
K4	Cutout XT60 J12, corners R1.5	16.5 × 9.4	THRU	2 wall 2	148.40	23.80	
K5	Cutout XT30 J13, corners R1.5	11 × 6	THRU	2 wall 2	115.50	22.10	
L1	EXP plate screw	M3	5	2 wall 2	235.50	30.00	tap drill 2.5, max 5.5 deep (wall 6)
L2	EXP plate screw	M3	5	2 wall 2	192.50	30.00	tap drill 2.5, max 5.5 deep (wall 6)

X, Y: from the datum 0,0 of the view named (lower-left outer corner as drawn). Depth = full thread depth; drill depth limits in the note column.

Check: 76 of 76 tapped holes match the thread data of rf-case.solidcraft (axis, size and depth).

NOTES

1 Master geometry: rf-case.step (Tray, Lid, EXP plate). This drawing adds threads, tolerances and finish.

2 Material EN AW-6061-T6 (EN AW-6061-T6 accepted) for tray, lid and EXP plate.

3 Finish: conductive conversion coating (JLCCNC: Conductive anodizing, natural), MIL-DTL-5541 Type II Class 3 equivalent, all surfaces. No insulating anodising.

4 General tolerances ISO 2768-mK. Hole positions ±0.1 for E, F, H, J (connector flanges).

5 Threads ISO metric coarse 6H. Blind holes: keep the max drill depth (note column); never break through into the pocket or the outside face.

6 Internal radii: pocket R10, cutouts R1.5, concave edges at lugs/bosses R ≤ 2.

7 Break all sharp edges 0.2 to 0.4, except the lid seat (flat, sharp, conductive).

8 Engrave the lid 0.5 deep as modelled (Arial bold).

9 Deburr the SO-239, SMA and cutout edges inside and outside (RF and cable contact).

PARTS PER UNIT

Tray 1, lid 1, EXP plate 1 (CNC, this drawing)

Printed spacer 4, printed washer 4 (sheet 5)

Screw M3 × 10 countersunk ISO 10642 A2: 16 (lid)

Screw M3 × 6 button/cheese head A2: 43 (28 SO-239, 7 GND boss, 2 RF board on S3 and S6, 4 controller board, 2 EXP plate)

Screw M5 × 16 hex socket ISO 4762 A4: 5 (4 mounting, 1 earth strap)

Screw M2 × 4 cheese head A2: 12 (SMA flanges)

Standoff M3 hex 5.5 AF, 10 mm female-male: 6 (pads S1..S6)

Standoff M3 hex 5.5 AF, 20 mm male-female: 4 (S1, S2, S4, S5, controller)

Connector Amphenol RF 83-798 SO-239: 7. Amphenol RF 901-9215 SMA: 3

Tinned copper braid with M5 lug: 1 (earth, to the RVS earth bar)

Shack Relay Matrix rev C - RF unit case	
Hole table, notes, parts list	Dwg RF-CASE-01 rev A
Material EN AW-6061-T6 (6082-T6 accepted). Finish: conductive conversion coating (JLC 'Conductive anodizing'), MIL-DTL-5541 Type II Cl 3.	Sheet 7 / 7
Tolerances ISO 2768-mK. Dimensions in mm. Third angle projection. Master geometry: rf-case.step	2026-10-24 PE1RGU (P. Doornekamp)